CONSTRUCTING STRATUM CONTOURS ON REAL MAPS
(Marlborough sheet)

The accompanying map is a reproduction of part of the BGS 1:50000 map of the area around Marlborough. It shows the topographic contours and the lower and upper boundaries of the Middle Chalk. The aim of this exercise is to use stratum contours to determine and illustrate the orientation of those two boundaries and to find the thickness of the Middle Chalk. Note that whereas perfectly planar surfaces have stratum contours which are straight, parallel, and equidistant, the surfaces encountered in nature are rarely perfectly planar and so have stratum contours which are only approximately straight, parallel, and equidistant.

1. Using a coloured pencil (mark the colour in the key), draw stratum contours on the upper boundary of the Middle Chalk:
   • Mark all the points where the 225 m topographic contour crosses the Upper Chalk / Middle Chalk boundary.
   • Now join all the marked points, remembering the following rules:
   − the stratum contour can only cross the boundary where the 225 m topographic contour also crosses it;
   − the stratum contour can only cross the 225 m topographic contour where the boundary also crosses it;
   − the statum contour should be as straight / gently curved as possible (there should be no sharp kinks).
   • Repeat the above for the 240 m, 255 m, 210 m, 195 m, 180 m, 165 m, 150 m, and 135 m contours.
   Attempt to keep the stratum contours as parallel and equidistant as possible. After drawing each contour, label it with its height.

2. Using a different coloured pencil (again mark the colour in the key) on a second copy of the map, draw stratum contours on the lower boundary of the Middle Chalk. These should have similar (but not necessarily identical) trends to the stratum contours on the upper boundary. Draw the contours for 135 m, 150 m, 165 m, 180 m, 195 m, and 210 m.

3. Construct a vertical cross section along line A-B showing the surface topography and the upper and lower boundaries of the Middle Chalk. Use the same horizontal scale as on the map and a vertical scale of 1:10000 − what vertical exaggeration does this lead to? The upper and lower boundaries should be extended above the topographic surface (as dashed lines) as well as below it. Complete your cross-section by including on it:
   (a) a title which states the orientation (strike/dip) of the section;
   (b) labels indicating which end of the section is north-west and which south-east;
   (c) a vertical and horizontal scale;
   (d) a stratigraphic column drawn to scale containing a colour and letter key to each unit;
   (e) colour shading and letter key for the lithological units on the section itself.

4. What is the strike and dip of the Lower / Middle Chalk boundary at locality C on the map? (NB. North is oblique to the edges of the paper.)

5. What is the thickness (to the nearest 10 m) of the Middle Chalk at this point? This thickness varies across the map − how do we know that?